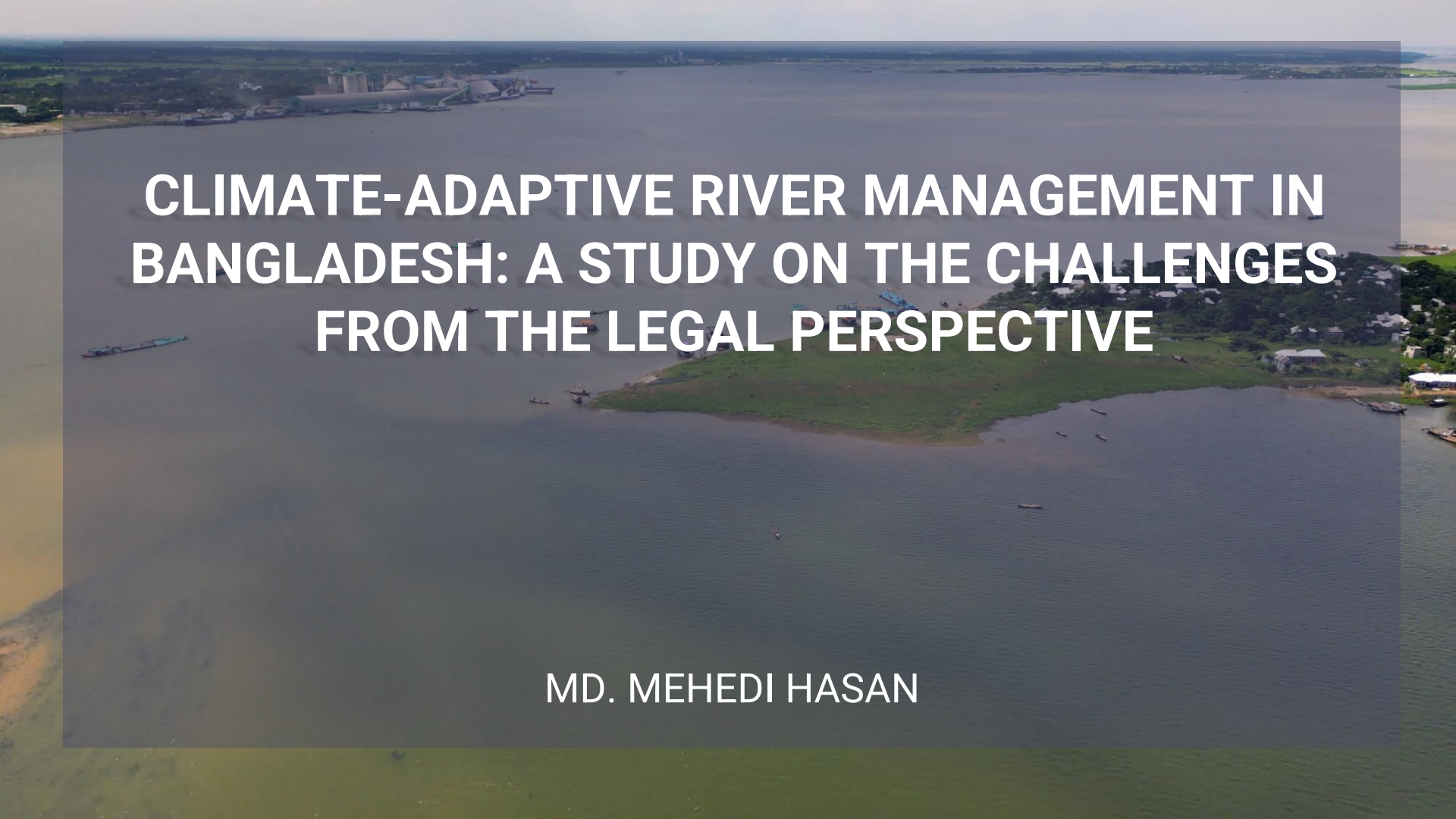


CLIMATE-ADAPTIVE RIVER MANAGEMENT IN BANGLADESH: A STUDY ON THE CHALLENGES FROM THE LEGAL PERSPECTIVE

MD. MEHEDI HASAN

Background

01

Bangladesh's Climate Challenges

- Ganges-Brahmaputra-Meghna basin
- Floods, Droughts, Salinity Intrusion, and Cyclones
- River Ecosystems and Dependent Communities

02

Existing Governance Issues

- Bangladesh Water Act (2013)
- National Water Policy (1999)
- National River Conservation Commission Act (2013)

03

Need for Adaptive Strategies

- To Protect Ecosystems
- Ensure Water Resource Equity
- Mitigate impacts on Vulnerable Communities

04

Integration with Global Efforts

- Align with sustainable development goals
- Incorporate international frameworks
- Bridge policy and environmental justice gaps

Research Questions

Question #1

How does Bangladesh’s legal framework address climate-adaptive river management?

Question #2

What are the legal challenges of climate-adaptive river management in Bangladesh?

Question #3

What legal measures can be taken to strengthen climate-adaptive river management in Bangladesh?



Research Objectives

Objective #1

To examine Bangladesh's legal framework for addressing climate-adaptive river management.

Objective #2

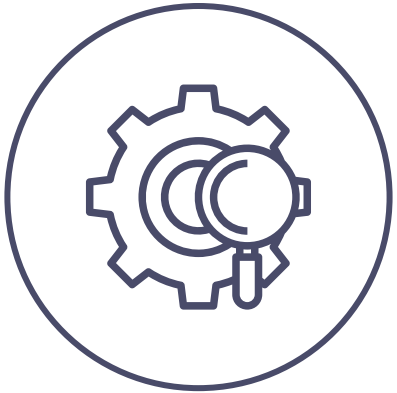
To identify key legal challenges to climate-adaptive river management in Bangladesh.

Objective #3

To recommend improvements to enhance climate resilience in Bangladesh's river management laws.



Methodology



Qualitative Approach

Used secondary data for analysis.



Literature Review

Analyzed existing studies on legal frameworks and climate resilience



Thematic Analysis

Identified recurring regulatory and governance issues.



Rights-Based Focus

Highlighted socio-economic vulnerabilities and equity in governance.

Literature review

Sium et al. (2023)

Analyze river morphology changes due to erosion and sedimentation but note the lack of geospatial data integration into long-term governance frameworks.

Hossain and Islam (2023)

Discuss institutional overlaps in water governance but highlight the absence of a centralized regulatory body for coordination and accountability.

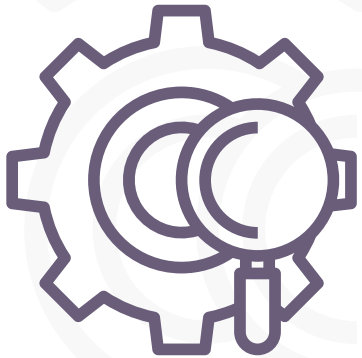
Rahman et al. (2022)

Link salinity intrusion to climate change impacts but identify fragmented policy responses and limited scaling of community-driven adaptive strategies.

Islam et al. (2021)

Examine socio-economic impacts of riverbank erosion but point out the absence of resettlement policies and participatory governance for displaced communities.

Scope



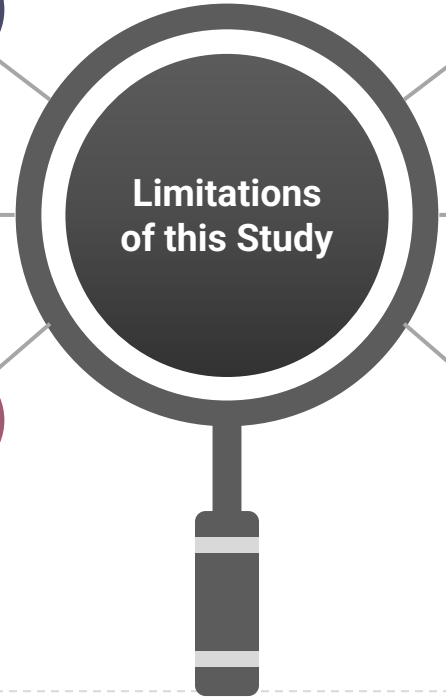
Inclusion

- Legal and policy review for climate adaptation.
- Adaptive practices in GBM river systems.
- Global best practices and reforms.

Exclusion

- Field surveys or primary data collection.
- Detailed hydrological or climate modeling.
- Broader geopolitical or economic evaluations.

Limitations



01

Reliance on Secondary Data

The study primarily uses secondary sources, limiting direct insights from stakeholders.

02

No Primary Data Collection

No field interviews or community surveys were conducted.

03

Absence of Field Validation

Findings were not validated through fieldwork or stakeholder engagement..

04

Scope Restriction

Focus was limited to specific legal and policy frameworks, excluding broader environmental factors.

05

Narrow Focus

Study emphasized legal and policy frameworks, excluding detailed ecological and socio-economic assessments.

06

Time Constraints

Limited timeframe restricted the scope of exploration and the depth of analysis.

Findings

1. **Gaps in Legal Frameworks**
   Existing laws like the Bangladesh Water Act (2013) and National Water Policy (1999) lack explicit climate resilience provisions.

2. **Lack of Policy Integration**
   Fragmentation between climate adaptation strategies (e.g., BCCSAP, Delta Plan 2100) and national water policies hinders coherent river management.

3. **Institutional Challenges**
   Overlapping responsibilities among agencies and limited resources weaken enforcement and adaptation efforts.

4. **Transboundary Water Issues**
   Absence of climate resilience clauses in agreements with upstream countries exacerbates water scarcity during the dry season.

5. **River System Vulnerabilities**
   Prone to flooding, droughts, salinity intrusion, and cyclones, impacting ecosystems and livelihoods.

6. **Socio-Economic Impacts**
   Policies fail to address vulnerabilities of river-dependent communities, limiting equitable access and participation.

7. **Community-Based Adaptation**
   Local practices exist but lack institutional support for broader implementation.

8. **Learning from Global Practices**
   International frameworks and country-specific strategies provide actionable models for adaptive governance in Bangladesh.

Recommendation

Integrate Climate Resilience into Policies

Enhance the Bangladesh Water Act (2013) and National Water Policy (1999) to incorporate adaptive measures, aligning with national strategies like BCCSAP and Delta Plan 2100.

Improve Institutional Coordination

Promote collaboration among agencies through an integrated water management framework, supported by capacity-building and dedicated resources.

Empower Community-Based Adaptation

Strengthen grassroots resilience by engaging local communities in water governance and providing tools, training, and resources for sustainable management.

Operationalize Legal Rights for Rivers

Implement river rights established by the Turag River case through enhanced legislation, community stewardship, and digital monitoring systems.

Enhance Transboundary Cooperation

Foster collaborative agreements with upstream countries, inspired by successful models like the Mekong River Commission, for equitable and adaptive water sharing.

Advance Disaster Preparedness

Invest in cutting-edge early warning systems and public education to mitigate risks from floods and cyclones, leveraging global expertise and partnerships.

Conclusions



Bangladesh's rivers face severe climate impacts, but existing laws lack climate resilience provisions. Institutional inefficiencies and fragmented governance further hinder adaptation, highlighting the need for legal reforms, better coordination, and community engagement.



QUESTIONS

THANK YOU !

MD. MEHEDI HASAN